

Jordan arc if the mapping is one-to-one. If distinct points of $[a, b]$ are mapped onto distinct points of $\bar{\mathbb{C}}$ and the image of b is the same as that of a , then γ is said to be a *simple, closed curve* or a *Jordan curve*. In other words, a Jordan curve is a homeomorphism in $\bar{\mathbb{C}}$ of the unit circle. The *Jordan curve theorem* (Heins 1962, p. 155) says that the complement of the trace of a Jordan curve γ with respect to $\bar{\mathbb{C}}$ has precisely two components. One of these two components is bounded in the Euclidean metric if the trace of γ lies in $\mathbb{C}$. That component is called the *inside* of γ , while the other component is the *outside*. A Jordan curve $\gamma : [a, b] \rightarrow \bar{\mathbb{C}}$ is said to be *positively oriented* if the inside of the curve lies on the left of the moving point $\gamma(t)$ as t increases from a to b .

An arc $\gamma : [a, b] \rightarrow \mathbb{C}$ is said to be *rectifiable* if it has finite length. This means that, for some finite number L ,

$$\sum_{\nu=1}^n |\gamma(t_\nu) - \gamma(t_{\nu-1})| \leq L$$

for every partition $\{a = t_0 < t_1 < \dots < t_n = b\}$.

An arc $\gamma : [a, b] \rightarrow \mathbb{C}$ is *piecewise continuously differentiable* if there exists a partition $\{a = t_0 < t_1 < \dots < t_n = b\}$ such that, in each of the intervals $[t_{\nu-1}, t_\nu]$ for $\nu = 1, \dots, n$, the functions $\Re \gamma$ and $\Im \gamma$ are continuously differentiable, having one-sided derivatives at the endpoints. Such an arc is rectifiable, and

$$\int_a^b |\gamma'(t)| dt$$

gives its *length*.

An arc $\gamma : [a, b] \rightarrow \mathbb{C}$ is said to be *analytic* if, about each point $t_0 \in (a, b)$, the function γ can be expanded in a power series

$$\gamma(t) = c_0 + c_1(t - t_0) + \dots \quad (c_1 \neq 0)$$

which converges in some interval $|t - t_0| < \delta$.

1.1 THE FUNDAMENTAL THEOREM OF ALGEBRA

In this book, a polynomial

$$f(z) = a_0 + a_1 z + \dots + a_{n-1} z^{n-1} + a_n z^n \quad (1.1.1)$$

is always supposed to have complex coefficients $a_0, \dots, a_n$ unless otherwise indicated. If $a_n \neq 0$, then the polynomial f is of *degree* n . This is written as $\deg f = n$, and a_n is called the *leading coefficient*. A polynomial with leading coefficient 1 is said to be *monic*.

An important relation between two polynomials f and g , with $\deg f = n$ and $\deg g = m \leq n$, is the *division transformation*. It guarantees a representation

$$f(z) = q(z)g(z) - r(z) \quad (1.1.2)$$

where q is a polynomial of degree $n - m$ and r is one of degree less than m . A constructive verification of (1.1.2) may be carried out by equating coefficients or

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by a division scheme. Repeating this process with the pair f, g replaced by g, r , etc., we arrive at the *Euclidean algorithm*. It may formally be described by

$$\begin{aligned} f_0 &:= f(z), & f_1(z) &:= g(z), \\ f_j(z) &:= q_j(z)f_{j+1}(z) - f_{j+2}(z) & (j = 0, 1, \dots). \end{aligned} \quad (1.1.3)$$

Since the degrees of the polynomials f_j are strictly decreasing, we finish with an equation $f_{k-1}(z) = q_{k-1}(z)f_k(z)$, where $f_k(z) \neq 0$. We may also allow the possibility that $\deg g > \deg f$. Then (1.1.2) holds with $q(z)$ being identically zero and $r(z) = -f(z)$, that is, $f_2 = -f$. Hence, except for a sign, the first step of the Euclidean algorithm interchanges f and g . The relations (1.1.3) show that any common divisor of f_0 and f_1 is a divisor of all the successors $f_2, \dots, f_k$. Conversely, f_k is a divisor of all its predecessors $f_{k-1}, \dots, f_0$. Hence f_k is a *greatest common divisor* of f and g . Moreover, we have factorizations

$$f(z) = f_k(z)\varphi(z) \quad \text{and} \quad g(z) = f_k(z)\psi(z),$$

where φ and ψ are appropriate polynomials.

A point $\zeta \in \mathbb{C}$ is a *zero of multiplicity* m , or of *order* m , or an *m-fold zero* of f if

$$f(\zeta) = f'(\zeta) = \dots = f^{(m-1)}(\zeta) = 0 \quad \text{and} \quad f^{(m)}(\zeta) \neq 0.$$

A zero of multiplicity 1 is said to be *simple*. If $f(\zeta) \neq 0$, then we may consider ζ as a zero of multiplicity 0. In the case where $m \geq 1$, the division transformation with $g(z) := (z - \zeta)^m$ yields that $f(z) = q(z)(z - \zeta)^m$. In fact, if the polynomial r in (1.1.2) were not identically zero, then either r itself or one of its derivatives of order less than m would be a non-vanishing constant. However, $r^{(\mu)}(\zeta) = 0$ for $\mu = 0, \dots, m - 1$; this is a contradiction.

In this way we also conclude that a polynomial of degree $n \geq 1$ cannot have more than n zeros, counted according to their multiplicities. However, it is a non-trivial question whether an arbitrary polynomial of positive degree has a zero at all. Its answer is crucial for the whole book, and is the main result of this section.

Theorem 1.1.1 (Fundamental theorem of algebra) *Every polynomial of positive degree has a complex zero.*

Proof Let f be a polynomial of positive degree. Then $|f(z)|$ tends to infinity as $|z| \rightarrow \infty$. Hence there exists an $R > 0$ such that $|f(z)| > |f(0)|$ for $|z| > R$. As a continuous function, $|f(\cdot)|$ attains a minimum on the compact disc $\bar{\mathcal{D}}(0; R)$. If this occurs at ζ , then $|f(\zeta)| \leq |f(0)|$, and so $|f(\zeta)| = \min_{z \in \mathbb{C}} |f(z)|$.

Now, assuming that f has no zero in $\mathbb{C}$, we may write

$$f(\zeta + z) = a + bz^k + z^{k+1}g(z)$$

with some integer $k \geq 1$, coefficients $a, b \in \mathbb{C} \setminus \{0\}$, and a polynomial $g(z)$. Corresponding to ω , chosen as a k -th root of $-a/b$, there exists a $t \in (0, 1)$ so that $t|\omega^{k+1}g(t\omega)| < |a|$. For this pair t, ω , we obtain

$$\begin{aligned} |f(\zeta + t\omega)| &= |a + b(t\omega)^k + (t\omega)^{k+1}g(t\omega)| = |a(1 - t^k) + (t\omega^{k+1}g(t\omega))t^k| \\ &< |a|(1 - t^k) + |a|t^k = |a| = |f(\zeta)|. \end{aligned}$$

This is a contradiction. $\square$

We already know from the division transformation that, if ζ is a zero of f , then $f(z) = (z - \zeta)f_1(z)$, where $\deg f_1 = \deg f - 1$. Repeating this reasoning, we may restate Theorem 1.1.1 in the following refined form.

Theorem 1.1.2 Every polynomial f of positive degree n has a total of n complex zeros, and may be represented as

$$f(z) = a_n \prod_{j=1}^k (z - z_j)^{m_j},$$

with $a_n \neq 0$, distinct zeros $z_1, \dots, z_k$, and positive integers $m_1, \dots, m_k$ satisfying $m_1 + \dots + m_k = n$. $\square$

In view of this theorem, the following problems arise for a polynomial f .

Problem 1 Given a subset Ω of the complex plane, how many zeros of f are contained in Ω ?

Problem 2 Given an integer m , with $1 \leq m \leq \deg f$, determine a subset of $\mathbb{C}$ which contains at least m zeros of f .

A comprehensive study of these problems will follow in Part II. We anticipate two results for immediate use.

An answer to Problem 1 is contained in the following theorem. We state it in its most general form, where it holds for meromorphic functions, but prove it only for rational functions, which clearly include the case of polynomials.

Theorem 1.1.3 (Principle of the argument) Let γ be a positively oriented Jordan curve in the complex plane, and let f be a meromorphic function defined in a domain which contains γ together with its inside. Suppose that f has neither a pole nor a zero on γ . Let z traverse γ once in the positive direction. Then the net change of $\arg f(z)$ is equal to $2\pi(m - \ell)$, where m is the number of zeros and ℓ is the number of poles lying inside γ and these are counted according to their multiplicities.

Proof for rational functions. Take a starting point a on γ and select a determination of $\arg(z - \zeta)$ in a small neighbourhood of a . Then, as z goes around γ once in the positive direction, the net change in the argument of $z - \zeta$ increases by 2π if ζ lies inside γ , but is zero if ζ lies outside.

A rational function can be written as $f = P/Q$, where P and Q are polynomials without a common zero. Then $\arg f(z) = \arg P(z) - \arg Q(z)$. Omitting the trivial case where P is a constant, we obtain from Theorem 1.1.2 a representation $P(z) = a_n \prod_{j=1}^k (z - z_j)^{m_j}$, which implies that

$$\arg P(z) = \arg a_n + \sum_{j=1}^k m_j \arg(z - z_j).$$

A corresponding formula holds for $\arg Q(z)$. With this and the preceding observation, we readily arrive at the desired result. $\square$

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Suppose that a polynomial has m zeros of modulus not exceeding r . Then, in a geometrical interpretation, the disc $\overline{D}(0; r)$ contains m zeros. Therefore, the following theorem gives a solution of the previous Problem 2.

Theorem 1.1.4 Let $f(z) = \sum_{\nu=0}^n a_\nu z^\nu$ be a monic polynomial of degree n , and let $m \in \mathbb{N}$ with $m \leq n$. Then f has at least m zeros of modulus not exceeding

$$2 \max_{0 \leq \mu \leq m-1} |a_\mu|^{1/(n-\mu)}.$$

Proof We first prove a weaker statement. Let f belong to the class $\mathcal{M}_{m,n}$ consisting of those monic polynomials $\sum_{\nu=0}^n a_\nu z^\nu$ of degree n for which $|a_\mu| \leq 1$, where $\mu = 0, \dots, m-1$. Arranging the zeros z_ν of f as $|z_1| \leq \dots \leq |z_n|$, we assert that $|z_m| \leq 2$.

To verify this, we proceed as follows. Suppose that $|z_{m+1}| > 2$, which is the only non-trivial case. We want to factor out the zeros $z_{m+1}, \dots, z_n$. Let ζ be one of them, and define

$$g(z) := \sum_{\nu=0}^{n-1} b_\nu z^\nu := \frac{f(z)}{z - \zeta} \quad \text{and} \quad b_{-1} := b_n := 0.$$

Then we find by equating coefficients that $a_\nu = -\zeta b_\nu + b_{\nu-1}$ for $\nu = 0, \dots, n$. From this, we conclude that

$$b_j = - \sum_{\nu=0}^j \zeta^{j-\nu+1} a_\nu \quad (j = 0, \dots, n-2),$$

and so, under our assumptions,

$$|b_\mu| \leq \sum_{\nu=0}^{\mu} \frac{|a_\nu|}{2^{\mu-\nu+1}} \leq \sum_{\ell=1}^{\mu+1} 2^{-\ell} < 1 \quad (\mu = 0, \dots, m-1).$$

Hence $g \in \mathcal{M}_{m,n-1}$. Repeating this process, we see that

$$h(z) := \sum_{\nu=0}^m c_\nu z^\nu := \frac{f(z)}{(z - z_{m+1}) \cdots (z - z_n)} \quad (c_m \equiv 1)$$

belongs to $\mathcal{M}_{m,m}$.

Next, let w be a zero of h . Then $w^m = -\sum_{\nu=0}^{m-1} c_\nu w^\nu$. Thus, in the case where $|w| > 1$, we have

$$|w| \leq \sum_{\nu=0}^{m-1} |c_\nu| \cdot |w|^{m-\nu} \leq \sum_{\ell=0}^{\infty} |w|^{-\ell} = \frac{1}{1 - |w|^{-1}},$$

which implies that $|w| \leq 2$. Hence, in any case, $|z_m| \leq 2$. The above assertion on the polynomials of the class $\mathcal{M}_{m,n}$ is verified.

The statement of the theorem is now easily deduced. Setting

$$\lambda := \max_{0 \leq \mu \leq m-1} |a_\mu|^{1/(n-\mu)}$$

and omitting a trivial case, we may suppose that $\lambda > 0$. Then $\phi(z) := \lambda^{-n} f(\lambda z)$ belongs to $\mathcal{M}_{m,n}$. Therefore, ϕ has at least m zeros of modulus not exceeding 2, and so f has at least m zeros of modulus not exceeding 2λ . $\square$

Let $f(z) = \sum_{\nu=0}^n a_\nu z^\nu$ and $g(z) = \sum_{\nu=0}^n b_\nu z^\nu = \prod_{\nu=1}^n (z - \omega_\nu)$ be two monic polynomials. Let ζ be a zero of f . Then

$$\prod_{\nu=1}^n (\zeta - \omega_\nu) = g(\zeta) = g(\zeta) - f(\zeta) = \sum_{\nu=0}^{n-1} (b_\nu - a_\nu) \zeta^\nu,$$

and so

$$\min_{1 \leq \nu \leq n} |\zeta - \omega_\nu| \leq \left(\sum_{\nu=0}^{n-1} |b_\nu - a_\nu| \cdot |\zeta|^\nu \right)^{1/n}.$$

Hence, to each zero ζ of f and any $\varepsilon > 0$, there exists a $\delta > 0$ such that every monic polynomial $g(z) = \sum_{\nu=0}^n b_\nu z^\nu$ satisfying $|a_\nu - b_\nu| < \delta$, for $\nu = 0, \dots, n-1$, has a zero ω with $|\zeta - \omega| < \varepsilon$. Hence each zero depends continuously on the coefficients.

Note that the multiplicity of the zero ζ did not play a role. However, in most of our applications we need the fact that, if ζ is of multiplicity m and the coefficients of g deviate only slightly from those of f , then g has not only one, but m zeros close to ζ . This is guaranteed by the following statement. For another result which can be interpreted as a continuity statement, see Theorem 5.5.1.

Theorem 1.3.1 (Continuity theorem) *Let*

$$f(z) = \sum_{\nu=0}^n a_\nu z^\nu = \prod_{j=1}^k (z - z_j)^{m_j} \quad (m_1 + \dots + m_k = n)$$

be a monic polynomial of degree n with distinct zeros $z_1, \dots, z_k$ of multiplicities $m_1, \dots, m_k$. Then, given a positive $\varepsilon < \min_{1 \leq j \leq k} |z_i - z_j|/2$, there exists a $\delta > 0$ so that any monic polynomial $g(z) = \sum_{\nu=0}^n b_\nu z^\nu$ whose coefficients satisfy $|b_\nu - a_\nu| < \delta$, for $\nu = 1, \dots, n-1$, has exactly m_j zeros in the disc

$$\mathcal{D}(z_j; \varepsilon) \quad (j = 1, \dots, k).$$

The proof will provide the following quantitative result in which the restriction on δ could be made more explicit (see Notes, § 1.8).

Supplement *In the situation of the preceding theorem, let*

$$A := \max \left\{ 1, 2|a_\nu|^{1/(n-\nu)} : \nu = 0, \dots, n-1 \right\},$$

and let the zeros of f be denoted by $\zeta_1, \dots, \zeta_n$, where an m -fold zero is now listed m times. Then, for sufficiently small $\delta > 0$, there exists a numbering of the zeros of g as $\omega_1, \dots, \omega_n$ such that $\max_{1 \leq \nu \leq n} |\omega_\nu - \zeta_\nu| \leq 4A\delta^{1/n}$.

Proof Expansions of the polynomials f and g at z_j yield

$$f(z + z_j) = \sum_{\nu=0}^n a_\nu^{[j]} z^\nu \quad \text{and} \quad g(z + z_j) =: g_j(z) = \sum_{\nu=0}^n b_\nu^{[j]} z^\nu,$$

where

$$a_\nu^{[j]} = \sum_{\kappa=\nu}^n \binom{\kappa}{\nu} a_\kappa z_j^{\kappa-\nu}, \quad b_\nu^{[j]} = \sum_{\kappa=\nu}^n \binom{\kappa}{\nu} b_\kappa z_j^{\kappa-\nu} \quad (\nu = 0, \dots, n-1),$$

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and $a_n^{[j]} = a_n = b_n = b_n^{[j]} = 1$. Furthermore, since z_j is a zero of f , of multiplicity m_j , we have $a_0^{[j]} = \dots = a_{m_j-1}^{[j]} = 0$, and so

$$b_\mu^{[j]} = b_\mu^{[j]} - a_\mu^{[j]} = \sum_{\kappa=\mu}^{n-1} \binom{\kappa}{\mu} (b_\kappa - a_\kappa) z_j^{\kappa-\mu} \quad (\mu = 0, \dots, m_j - 1).$$

Now, by applying Theorem 1.1.4 to g_j with $m = m_j$ and introducing

$$\rho_j := 2 \max_{0 \leq \mu \leq m_j-1} \left(\sum_{\kappa=\mu}^{n-1} \binom{\kappa}{\mu} |b_\kappa - a_\kappa| \cdot |z_j|^{\kappa-\mu} \right)^{1/(n-\mu)}, \quad (1.3.1)$$

we find that $\mathcal{D}_j := \{z \in \mathbb{C} : |z - z_j| \leq \rho_j\}$ contains at least m_j zeros of g . By choosing δ sufficiently small, the radii ρ_j can all be made smaller than ε . Then the discs $\mathcal{D}_1, \dots, \mathcal{D}_k$ are disjoint. With this, we conclude that $\mathcal{D}_j$ contains exactly m_j zeros of g .

To verify the supplement, it suffices to show that $4A\delta^{1/n}$ is an upper bound for the radii ρ_j in (1.3.1), at least for small $\delta > 0$. By Theorem 1.1.4, the moduli of the zeros of f are bounded by A . Since $A \geq 1$ and $\binom{\kappa}{\mu} < 2^\kappa$, we find that

$$\sum_{\kappa=\mu}^{n-1} \binom{\kappa}{\mu} |b_\kappa - a_\kappa| \cdot |z_j|^{\kappa-\mu} < 2^n \delta A^{n-\mu}.$$

Hence, for $0 < \delta < 2^{-n}$, we have

$$\rho_j < 2 \max_{0 \leq \mu \leq m_j-1} (2^n \delta)^{1/(n-\mu)} A \leq 4A\delta^{1/n}.$$

□

Remark 1.3.2 In Theorem 1.3.1, we found it convenient to consider only monic polynomials. If f is not monic, then its zeros depend continuously on a_ν/a_n for $\nu = 0, \dots, n-1$. The coefficient a_n may also vary provided that it does not approach the origin. If $a_n \rightarrow 0$ while the other coefficients remain fixed, then at least one zero tends to infinity. This is seen, with the help of Theorem 1.1.4, by considering the polynomial $h(z) := z^n f(1/z)$, whose zeros are the reciprocals of those of f . Then we still have continuity for the zeros of f if we consider them as points on the Riemann sphere with the standard topology, or equivalently, as points in the extended complex plane $\widehat{\mathbb{C}}$ with the chordal metric.

Remark 1.3.3 By a change of the basis from $1, z, \dots, z^n$ to $Q_0(z), \dots, Q_n(z)$, as discussed in the preceding section, we can conclude that the zeros of a polynomial $\sum_{\nu=0}^n c_\nu Q_\nu(z)$ depend continuously on $c_0, \dots, c_n$ provided that these coefficients vary such that the degree is not reduced.

Remark 1.3.4 In view of the supplement, we may say that the zeros of a polynomial of degree n , as functions of the coefficients, satisfy a Lipschitz condition of order $1/n$.

Restricted to the case of polynomials, various important results of complex analysis can be deduced from Theorem 1.3.1. An example would be the maximum modulus principle. Here we prefer a variant useful for the comparison of two polynomials.

This permits us to determine the signature. We first introduce some notation. Let $a_0, \dots, a_n$ be a sequence of non-vanishing real numbers. Then we count a variation of sign at the index k , where $1 \leq k \leq n$, if $a_{k-1}a_k < 0$, and denote by $V(a_0, \dots, a_n)$ the total number of variations of sign in this sequence. For example, $V(-3, -2, 1, 5, -2) = 2$.

It is easily seen that $V(1, D_1, \dots, D_m)$ is the number of squares in (1.7.10) which have a negative coefficient. This leads us to the following result.

Theorem 1.7.14 (Jacobi) Let A be a symmetric matrix of rank m . Suppose that the principal minors $D_1, \dots, D_m$ of A are all different from zero. Then the quadratic form $Q(x) := x^T A x$ has signature $\sigma(Q) = m - 2V(1, D_1, \dots, D_m)$. $\square$

In the case where some of the principal minors $D_1, \dots, D_m$ vanish, attempts were made to define the counting function V in such a way that Theorem 1.7.14 remains valid. They were successful provided that not more than two consecutive principal minors vanish. If three or more consecutive principal minors are allowed to be zero, then appropriately constructed examples show that the validity of Theorem 1.7.14 cannot be preserved. However, it was discovered by Frobenius that for certain classes of matrices, which cover some important types, it is possible to establish an analogue of Theorem 1.7.14 without any restriction on the minors. We shall present one of his results, but only to the extent needed for our applications, see (Gantmacher 1986, § 10.10) for details. Let us start with some preparations.

A real Hankel matrix of order n is of the form

$$S := \begin{pmatrix} s_1 & s_2 & \dots & s_n \\ s_2 & s_3 & \dots & s_{n+1} \\ \vdots & \vdots & \ddots & \vdots \\ s_n & s_{n+1} & \dots & s_{2n-1} \end{pmatrix},$$

where $s_1, s_2, \dots, s_{2n-1}$ is a sequence of real numbers. We now extend the counting function V to vanishing intermediate elements.

Definition 1.7.15 Let $a_0, \dots, a_n$ be a sequence of real numbers such that $a_0 a_n \neq 0$. For $\nu = 0, \dots, n$, define

$$a_\nu^* := \begin{cases} a_\nu & \text{if } a_\nu \neq 0, \\ (-1)^{\nu(j-1)/2} a_{\nu-j} & \text{if } a_\nu = \dots = a_{\nu-j+1} = 0, \quad a_{\nu-j} \neq 0, \quad j \in \mathbb{N}. \end{cases}$$

Then $V^*(a_0, \dots, a_n) := V(a_0^*, \dots, a_n^*)$, which is the number of variations of sign in the sequence of non-vanishing numbers $a_0^*, \dots, a_n^*$.

Note that this extension of V to vanishing elements is different from that given in Chapter 10 (see Definition 10.1.1).

Theorem 1.7.16 (Frobenius) Let S be a real Hankel matrix of rank m . Denote by $D_1, \dots, D_m$ the principal minors of S , and suppose that $D_m \neq 0$. Then the quadratic form Q defined by S has signature $\sigma(Q) = m - 2V^*(1, D_1, \dots, D_m)$. $\square$

Remark 1.7.17 The investigation of Frobenius shows that, if we are in the situation of Theorem 1.7.16 and we have

$$D_k \neq 0, \quad D_{k+1} = \dots = D_{k+\ell} = 0, \quad D_{k+\ell+1} \neq 0,$$

and ℓ is odd, then $\text{sgn}(D_k D_{k+\ell+1}) = (-1)^{(\ell+1)/2}$. This implies that $r(Q) = \sigma(Q)$, or equivalently, that $V^*(1, D_1, \dots, D_m) = 0$, if and only if the principal minors $D_1, \dots, D_m$ are all positive.

Remark 1.7.18 In its full generality, the result of Frobenius also covers the case where $D_m = 0$, but, in our later applications, this situation will not occur.

The complex analogue of real quadratic forms are Hermitian forms, see (Mirsky 1955, § 12.6). A Hermitian form in n complex variables $z_1, \dots, z_n$ is defined by

$$F(z) := z^* H z = \sum_{\mu=1}^n \sum_{\nu=1}^n h_{\mu\nu} \bar{z}_\mu z_\nu, \quad (1.7.11)$$

where $H = (h_{\mu\nu}) \in \mathbb{C}^{n \times n}$ is a Hermitian matrix. Admissible transformations are defined as in the real case, except that the corresponding matrix C may have complex elements. There always exists an admissible transformation which transforms F into

$$\tilde{F}(\zeta) = \sum_{\nu=1}^{n_1} |\zeta_\nu|^2 - \sum_{\nu=1}^{n_2} |\zeta_{n_1+\nu}|^2 \quad (n_1 + n_2 \leq n).$$

Again, $r(F) := n_1 + n_2$ is defined as the rank, and $\sigma(F) := n_1 - n_2$ as the signature of F . Most of the results on Hermitian forms are strictly analogous to those on real quadratic forms. We shall need only the following statement, which corresponds to Theorem 1.7.14.

Theorem 1.7.19 (Hermite) Let H be a Hermitian matrix of rank m . Suppose that the principal minors $D_1, \dots, D_m$ of H are all different from zero. Then the Hermitian form $F(z) := z^* H z$ has signature $\sigma(F) = m - 2V(1, D_1, \dots, D_m)$. $\square$

Clearly, Theorem 1.7.16 cannot have a reasonable analogue in the Hermitian case.

1.8 NOTES

Section 1.1

The Euclidean algorithm is named in appreciation of an analogous division scheme used by Euclid (about 300 BC) for the determination of the greatest common divisor of two integers. Descartes observed in 1637 that, if ζ is a zero of $f(z)$, then $z - \zeta$ factors out; probably, this was already known to Thomas Harriot (about 1560–1621).

Possibly the first approach to the fundamental theorem of algebra was made by Peter Roth when he mentioned in his *Arithmetica philosophica* (Nuremberg, 1608) that an algebraic equation can have as many solutions as its degree, but not more. He added (freely translated from German) that this does not mean that every cubic, biquadratic, quintic, etc. equation must have three, four, five, etc. solutions, but that the maximum number of solutions is equal to the degree. We do not know if he had thought of

unsolvable equations, or only on a reduction of the number of *distinct* solutions due to multiple roots.

The next step was made by A. Girard in his *L'invention nouvelle en l'algebre* of 1629 where he stated without a proof: 'Toutes les equations d'algebre reçoivent autant de solutions, que la denomination de la plus haute quantité le demontre, excepté les incomplettes.' Later, he did not mention exceptions any more, but gave explanations which amount to the observation that a solution can be real, complex, or 'something similar'.

First attempts for a proof of the fundamental theorem of algebra were made by d'Alembert in 1746, Euler in 1749, Lagrange in 1772, and Laplace in 1795. Gauss criticized all of them. D'Alembert came to a situation where he had to assume the existence of a minimum; it would now follow by a compactness argument, which was not available in his time. The major objection against the three other proofs was that their authors already assumed the existence of some obscure solution (as Girard had postulated) and proved that it must be a complex number. Again, these days, this cannot be seen as a severe problem since now it is not difficult to show by purely algebraic means that for every polynomial with coefficients in a field there exists a splitting field. Thus algebra guarantees that a polynomial with complex coefficients has certain 'generalized' zeros, while the fundamental theorem of algebra states that they are again complex numbers, or equivalently, that the *field of complex numbers is algebraically closed*.

It should be emphasized that, despite its name, the fundamental theorem of algebra is not a purely algebraic theorem. Since the complex numbers form a topological field, proofs of that theorem cannot avoid analytical, topological, or geometrical tools completely, although they may sometimes be quite hidden.

Gauss himself gave four proofs, one in 1799, two in 1816, and one in 1849; see (Gauss 1876, pp. 1-30, 31-56, 57-64, 71-102). The first proof is contained in his doctoral dissertation, written at the age of 22. He considered polynomials $f(z)$ with real coefficients and justified, by delicate geometric arguments, that the curves $\Re f(z) = 0$ and $\Im f(z) = 0$ intersect. His fourth proof is of a similar nature, but he allowed complex coefficients. Analyzing these two proofs, Ostrowski in 1920 pointed out some possible gaps and presented the desirable supplements in a rigorous form, see (Gauss 1981, Part 2, article 3, pp. 1-18; Ostrowski 1983, pp. 538-53). An incontestable variant of Gauss's first proof was also given by Motzkin and Ostrowski (1933).

Gauss's second proof is strongly algebraic. He adopted ideas of Euler, but did calculations in terms of an unknown instead of assuming the existence of a 'generalized' zero. It is in a similar way that the existence of a splitting field is proved these days. His third proof is purely analytic, and is related to a familiar proof in complex analysis using a contour integral of f'/f .

For more details and references on the history of the fundamental theorem of algebra, see (Ebbinghaus *et al.* 1983, Chapter 4; Fine and Rosenberger 1997). Up to now, an overwhelming number of proofs have appeared. We mention only a few, classifying them according to their authors' major intention.

Elementary proofs are those which do not need more than standard calculus. The above-mentioned proof of d'Alembert is of that type. It was simplified and improved by Argand (1814) and Cauchy (1820). More modern versions are due to Fefferman (1967) and Terkelsen (1976), whose proof is the one presented in this book. Other elementary proofs were obtained by Redheffer (1957, 1964) and Wolfenstein (1967).

Complex analysis proofs are amongst the shortest and most familiar ones. Gauss's third proof belongs to this class. Bieberbach (1952) presents five different proofs: one by Ankeny (1947), who used Cauchy's integral theorem directly; a proof by van der Waerden based on the maximum modulus principle; the familiar proofs using the theorems of Liouville and Rouché; and Cauchy's proof, which Bieberbach based on the open mapping theorem. Ankeny's proof was modified by Boas (1964). The principle of the argument and a related formula for the contour integral of a logarithmic derivative lead to further proofs.

Predominantly algebraic proofs. The most prominent representatives are the proof of Laplace in 1795 and the second proof of Gauss. The ingenuity of Laplace's proof in a completed form is exhibited in (Ebbinghaus *et al.* 1983, pp. 95-7). It needs four tools, of which three are purely algebraic, while the only analytic tool is the intermediate-value theorem, used to guarantee that a polynomial of odd degree with real coefficients has a real zero. For references to further strongly algebraic proofs, see (Ebbinghaus *et al.* 1983).

Predominantly real proofs aim at avoiding complex numbers. Gauss's first proof is of that type. For other such contributions, see (Perron 1949), whose proof is also elementary, and the above-mentioned papers of Redheffer. By considering $f(z)f(\bar{z})$ instead of $f(z)$, it is enough (though not easier) to give a proof for polynomials with real coefficients.

Topological proofs. Arnold (1949) uses Brouwer's fixed point theorem, while Stein (1954) has a short proof based on elementary homology theory.

Constructive proofs were already proposed by Weierstrass in 1859 and 1891 (Weierstrass 1894, pp. 247-56; 1903, pp. 251-69). They have to satisfy further criteria for being acceptable as 'intuitionistic' (Brouwer and de Loor 1924; Weyl 1924, pp. 142-6; H. Kneser 1940; Rosenbloom 1945; M. Kneser 1981). H. W. Kuhn (1974) gave an unusual constructive proof based on a labeling procedure and a combinatorial lemma. In recent times, never-failing root-finding algorithms have gained a lot of attention, e.g., (Dejon and Nickel 1969; Hirsch and Smale 1979; Smale 1981; Ruscheweyh 1984). They can serve as constructive proofs provided that, after a careful study of their foundations, possible non-constructive arguments or assumptions concerning the existence of zeros have been eliminated.

Theorem 1.1.3, in its full generality, is proved in any book on complex analysis, e.g., (Ahlfors 1966, p. 151). Theorem 1.1.4 is due to Montel (1934/35).

Section 1.2

Symmetric polynomials and their applications can be found in books on algebra, e.g., (van der Waerden 1964). For a comprehensive classical treatment of material which has now fallen into oblivion, see (Serret 1878; Netto 1896; Weber 1898).

François Viète (1540-1603), also known under his Latin name Vieta, has been considered as a father of algebra. The formulae (1.2.2), for degrees n up to 5, follow from theorems contained in his book *De aequationum recognitione et emendatione*, published in 1615. It was clear to Girard in 1629 how to proceed for general n . The latter introduced the elementary symmetric polynomials, calling them 'première, deuxième, ..., dernière factions'. He also calculated power sums of zeros in terms of coefficients up to power 4. Newton, in his research on the theory of equations (1665-6), did the same for a polynomial of degree 8 up to power 8, and gave, in addition, the recurrence formula (1.2.9) for these cases, see (Newton 1967, p. 519).

The interpolation formula (1.2.5) was found by Waring in 1779, but has been named after Lagrange, who rediscovered it in 1812. Formula (1.2.6) was known to Gregory in 1670 and is contained in Newton's *Philosophiae naturalis principia mathematica* of 1687.

In (Ebbinghaus *et al.* 1983, p. 96), Theorem 1.2.2 is seen as a result of Newton from 1673. Waring proved it in his treatise *Miscellanea analytica de aequationibus algebraicis et curvarum proprietatibus* of 1762, and gave an alternative proof in his *Meditationes algebraicae* of 1770. It was also proved by Lagrange (1770, pp. 369-70).

Section 1.3

Continuity arguments for the zeros of a polynomial have often been used without a clear justification. The simple observation preceding Theorem 1.3.1 is taken from (van der Waerden 1973, p. 48). For more precise statements, some of them also being suitable for error estimates in computations, see (Weber 1898, pp. 148-53; Ostrowski 1940, Chapter 5, §16; H. Kneser 1942/43; Tutschke 1963; Zaubek 1963; Zedek 1965; Bhatia

et al. 1990). Theorem 1.3.1 and its supplement, which have been designed under the restriction of sparse preliminaries, include the result of Kneser and are essentially in accordance with that of Bhatia *et al.* when δ is small. These authors obtained a substantial improvement of Ostrowski's result. Like the latter, they prefer a somewhat different setting, where the roles of the polynomials f and g can be interchanged and no restriction on the size of δ is needed. For further results, see (Beauzamy 1999).

There are also results which show that, if the coefficients are differentiable functions of one or several parameters, then so are the zeros (Schumacher 1950; Brønštēin 1979).

Theorems 1.3.5 and 1.3.6, or variants, have been employed in many instances as simple consequences of the maximum modulus principle. The approach presented in this book allows us to gain the fundamental tools, needed for the study of polynomials, by elementary means.

Theorems 1.3.7 and 1.3.8, or variants, are proved in books on complex analysis, e.g., (Ahlfors 1966). A special form of Theorem 1.3.7 was first obtained by Rouché (1882). Hurwitz (1889) stated this result without mentioning Rouché, and used it for deriving a statement similar to Theorem 1.3.8. Conversely, Abian (1977) showed that Theorem 1.3.7 can be deduced from Theorem 1.3.8.

Section 1.4

For more material on orthogonal polynomials, we refer to (Freud 1971; Szegő 1975; Chihara 1978; Neval 1979).

It seems that the term *m-distribution* has been introduced in the just mentioned English translation of Freud's book. The m stands for *momenta*.

The result mentioned in Remark 1.4.5 has become famous under the name of Favard, although equivalent forms can be found in three prior books due to Perron, Wintner, and Stone; the first two appeared in 1929 and the last one in 1932; see (Szegő 1982, pp. 866–7).

The formula of Theorem 1.4.6 for $z \neq w$ was discovered by Christoffel (1858) in case of the Legendre polynomials and by Christoffel (1877) and Darboux (1878) for general orthogonal polynomials.

The Gaussian quadrature formula (Theorem 1.4.8), in the case where $\sigma(x) \equiv x$, originates from Gauss's Latin paper 'Methodus nova integralium valores per approximationem inveniendi', which appeared in September 1814, see (Gauss 1876, pp. 163–96). A simpler approach was found by Jacobi (1826). Christoffel (1858) discovered several representations for the coefficients ω_j ; for this reason, they are called *Christoffel numbers*. He also generalized the Gaussian quadrature formula by prescribing some of the nodes. Mehler (1864) extended the Gaussian quadrature formula to integrals with Jacobi weights, that is,

$$d\sigma(x) = (1-x)^\alpha(1+x)^\beta dx \quad (x \in (-1, 1), \alpha > -1, \beta > -1).$$

For general weight functions, the Gaussian quadrature formula is due to Christoffel (1877). The form (1.4.8) was initiated by Stieltjes's work on continued fractions in 1894; see (Gautschi 1981).

Section 1.5

The classical orthogonal polynomials have arisen in diverse branches of mathematics, such as the theory of continued fractions, mathematical physics, the study of extremal problems, and probability. They have their origins in the eighteenth and nineteenth century, but we cannot expect to find them in their familiar forms together with their typical properties in papers of that time. To some extent, it becomes a matter of interpretation to ascribe them to a certain author. In some cases, they have definitely not been named after the person who introduced them first (Szegő 1982, pp. 866–8).

A reference to the Jacobi polynomials is (Jacobi 1859). The Legendre polynomials were introduced by Legendre in a treatise on the attraction of ellipsoids in 1783, and were further explored by him in 1785.

The Chebyshev polynomials appeared as solutions of extremal problems investigated by the Russian mathematician P. L. Chebyshev (1821–94), who may be considered as the father of approximation theory. More than two dozen ways of spelling his name in Latin letters have been counted. Some typical other spellings used by English, French, and German writing authors are Chebichev, Tchebichef, Tchebicheff, Tchebychef, Tchebychev, Tschebyscheff, Tschebyschew, and Tschebyschow. Except for the last variant, a common phonetic transliteration is Čebyšev.

In his paper 'Théorie des mécanismes connus sous le nom de parallélogrammes' of 1854, Chebyshev studied the movement of the connecting rod of Watt's steam engine (Chebyshev 1961, pp. 109–43). This led him to the problem of the best approximation of a continuous function on an interval by polynomials of given degree. Theorem 1.5.4 is one of his results. The outstanding status of Chebyshev polynomials shows in numerous other theorems of approximation theory (Meinardus 1967; Rivlin 1990). Theorem 1.5.5, or variants, seem to have been discovered a number of times (Sós and Turán 1955; Schmeisser 1967b); for further references, see (Bhatia *et al.* 1990).

Two proofs of Theorem 1.5.7 were given by Koršine and Zolotareff (1873). Implicitly, the theorem is contained in Chebyshev's paper 'Sur l'interpolation dans le cas d'un grand nombre de données fournies par les observations' of 1859 (Chebyshev 1961, pp. 385–469).

For a discussion of the origins of the Laguerre and Hermite polynomials, we refer to (Szegő 1982, p. 866).

Section 1.7

For Theorems 1.7.1 and 1.7.11, see (Schur 1909). Grone *et al.* (1987) published a list of seventy conditions on a square matrix A , each of them equivalent to A being *normal*. The condition of Theorem 1.7.2 is amongst them. Three other conditions are the cases of equality in Theorem 1.7.11.

Theorem 1.7.3 for determinants with real elements was mentioned by Lord Kelvin to Muir in 1886. Muir (1902) published a proof. Another proof was given by E. J. Nanson. In the meantime, Hadamard (1893) had independently considered an extremal problem for determinants with complex elements; this led him to Theorem 1.7.3. All of these proofs are different from the one indicated in this book.

In the definition of norms, we considered only the case where $\mathbb{K} = \mathbb{C}$ since a vector norm $\|\cdot\|$, originally defined on $\mathbb{R}^n$, can be extended to $\mathbb{C}^n$ by

$$\|x + iy\| := \sup_{\theta \in \mathbb{R}} \|x \cos \theta - y \sin \theta\| \quad (x, y \in \mathbb{R}^n).$$

Theorem 1.7.5 is usually attributed to Gerschgorin (1931), but there are also other sources that deserve mention. It has been known for a long time (Lévy 1881; Desplanques 1887; Minkowski 1900; Hadamard 1903, pp. 13–14) that the determinant of a *diagonally dominant* $n \times n$ matrix A , that is,

$$|a_{\mu\mu}| > \sum_{\nu \neq \mu} |a_{\mu\nu}| \quad (\mu = 1, \dots, n),$$

does not vanish. In his lectures, Schur gave a short proof of this criterion for matrices of the form $A - zI$, referring to a remark of Frobenius, and mentioned its significance for eigenvalues. Schur's argument was communicated by Rohrbach (1931). Thus Schur and his pupils were aware of the first part of Theorem 1.7.5. Moreover, Gerschgorin (1931) and Rohrbach (1931) both used the transformation by a diagonal matrix described in Remark 1.7.7. It seems that in (Gerschgorin 1931) only the second statement of Theorem 1.7.5 was something really new. However, it was Gerschgorin who saw the importance of the theorem for numerical purposes and evoked interest in further investigations. See (Bodevig 1956, pp. 59–60; Brauer and Rohrbach 1969) for historical remarks, and (Taussky 1949) for a list of references.